(3) may within 30 days of receiving a copy of the request for a variance, submit to the chief inspector written comments on the request and provide a copy to the other two parties.

Postings

1.2.4 The manager shall ensure that a legible copy of each variance granted is posted and maintained on conspicuously located bulletin boards at the mine.

Authority to Enter a Mine

Authorization

1.3.1 Other than an inspector, only persons authorized by the manager shall enter or be permitted to enter a mine.

Posting

1.3.2 Notice to this effect shall be posted at all road entrances to the mine by the manager and, for non-operating mines, the contact information of a qualified person shall be included in the notice.

Unauthorized Access

1.3.3 Unless authorized by the manager, no persons shall enter or leave a mine except by a recognized means of entry or exit.

Posting and Distribution of *Mines Act*, Regulations, and Code

Manager's Responsibility

1.4.1 The manager must ensure that

(1) every employee is given a copy of the *Mines Act*, the regulations, and the code, and provided additional copies at cost, and

(2) a copy of the *Mines Act*, the regulations, and the code is maintained in every safety office and at other suitable locations and available to employees.

Hours of Employment

Hours of Work

1.5.1 Notwithstanding the provisions of the *Employment Standards Act*, the manager shall ensure that, over the period of any work schedule, no worker shall be scheduled to work more than an average of 50 hours per week.

(1) The manager shall not permit the employment of a person at a surface mine for a period longer than 16 hours in any 24 hours;

(2) The maximum scheduled shift underground in a mine shall be no longer than 10 hours, and the manager shall not permit the employment of a person underground at a mine for a period longer than 12 hours in a 24 hour period;

(3) 1.5.1 does not apply to emergency where life or property is in danger, or

(i) Where urgent work is essential to the continuation of the ordinary working of a mine, providing it is only on an infrequent basis, or

(ii) where work schedules incorporate periodic shift changes within a 24-hour period.

Occupational Health and Safety Committee

Committee Members

1.6.1 The manager shall establish and maintain a joint management-worker Occupational Health & Safety Committee

(a) at a mine where 20 or more workers are regularly employed, and

(b) at any other mine when ordered by an inspector.

1.6.2 The Occupational Health and Safety Committee shall be composed of management and an equal or greater number of worker representatives, and such committee shall have

(1) two or more worker representatives, chosen by the workers, and

(2) two co-chairpersons, one of the worker representatives and the other a management representative, and they may alternate chairing the meetings.

Inspection

1.6.3 The committee shall

(1) inspect as many of the work sites as it considers appropriate every month and as soon as possible after the inspection, meet to discuss its findings and any other matters concerning health and safety, and

(2) prepare minutes of the meeting including a description of conditions found during the inspection.

1.6.4 The minutes of the meeting referenced in section 1.6.3 (2) shall be signed by the OHSC co-chairpersons or their designates and a copy shall without delay

(1) be filed with the manager, and

(2) be forwarded to the local union(s), and

(3) be displayed in a conspicuous location at the mine until replaced by the minutes of the next meeting, and

(4) be made available to an inspector on request.

Accident Investigation

1.6.5 Repealed.

1.6.6 Repealed.

Cooperation With Committee

1.6.7 The manager and all persons working at the mine shall cooperate fully with the OHSC by

(1) providing it with every reasonable facility for carrying out its inspections and investigations, and

(2) allowing it access to all reports, plans, records and standards pertinent to the work of the OHSC, and

(3) correcting the safety hazards noted in the OHSC minutes by the date agreed by the OHSC committee.

OHSC Training

1.6.8 Where 20 or more workers are regularly employed, the manager shall arrange for a qualified person to provide the committee with training sessions on three occasions during the year.

Health and Safety Program

1.6.9 (1) The manager must ensure that a Mine Health and Safety Program is developed and implemented that includes the following sections

(a) a written policy statement,

(b) general safety rules,

(c) safe working procedures on a departmental basis,

(c.1) if emerging technology is used at the mine, safe working procedures for the emerging technology,

(d) a list of hazardous materials, safe handling procedures and antidotes,

(e) provision for the regular monthly crew safety meetings,

(f) procedures for accident and serious incident investigation,

(g) procedures for safety tour inspections,

(h) a written preventative training program, acceptable to the chief inspector, to educate OHSC members in the recognition, evaluation and prevention of adverse health effects resulting in Musculoskeletal Disorders (MSD) and in reporting related symptoms and injuries, and

(i) a program to address bullying and harassment, sexual harassment, and violence, which program includes

(i) establishing procedures for reporting and investigating complaints,

(ii) training supervisors and workers about the program,

(iii) a process to ensure that, after the training referred to in subparagraph (ii), supervisors and workers are kept aware of the program,

(iv) educating visitors about the program on entering the mine, and

(v) ensuring contractors are trained about the program on or before entering the mine.

(2) The manager must ensure that the OHSC

(a) reviews the Mine Health and Safety Program, including OHSC Training, for completeness and effectiveness on an ongoing basis, and

(b) submits its findings to the manager.

Entitlement to Time

1.6.10 The manager shall ensure that OHSC representatives are given reasonable time to carry out their duties as prescribed in the code and that time so spent is counted as time worked.

1.6.11 A worker health and safety representative chosen by the workers is required

(1) (a) at a mine where 9 but less than 20 workers are regularly employed, and

(b) at any other mine when ordered by an inspector.

(2) To the extent practicable, a worker health and safety representative has the same duties and functions as an OHSC.

1.6.12 Where there are 5 or less workers on a shift, or less than 9 workers in total employed at a mine, there shall be a crew safety meeting

(a) at the commencement of operations at a mine, and

(b) at least monthly thereafter, and

(c) minutes of these meetings shall be kept and made available to the inspector on request

Reportable Incidents

Reportable Incidents

1.7.1 (1) The following incidents are classified as dangerous occurrences and must be reported in accordance with section 1.7.2:

(a) an incident that causes serious injury or loss of life;

(b) an incident that had the potential to cause serious injury or loss of life.

(2) In addition to an incident referred to in subsection (1), the following incidents are classified as dangerous occurrences and must be reported in accordance with section 1.7.2:

(a) unexpected major groundfall, slope failure or subsidence;

(b) unexpected structural failure or collapse of any part of the mine;
(c) for dams or dikes:
  (i) unexpected cracking, subsidence, seepage, springs;
  (ii) loss of freeboard;
  (iii) washout, structural erosion, or any other deficiency that may adversely affect the integrity of the structure.
(d) equipment going out of control or an unexpected movement of equipment;
(e) unexpected inrush of water, mud, slurry or debris;
(f) unexpected significant inflow or release of explosive, flammable, toxic or other dangerous gas, or ignition or explosion of gas or dust;
(g) any of the following if it has the potential to cause, or causes, injury to persons or damage to equipment or property:
  (i) an incident involving explosives;
  (ii) outbreak of fire;
  (iii) an incident involving electrical equipment;
  (iv) failure of a pressure vessel or boiler;
  (v) unintentional contact with a falling object or drilling or other rotational equipment;
  (vi) exposure to hazardous environmental conditions;
(h) for underground mines,
  (i) unexpected cracking, seepage, failure or pressure on a bulkhead;
  (ii) an incident involving a mine hoisting plant including, without limitation, sheaves, hoisting rope, shaft conveyance, shaft, shaft timber or headframe structure,
  (iii) unplanned stoppage of the main underground ventilation system, or
  (iv) an underground fire.

Notification of a Reportable Incident

1.7.2 (1) In the event of a reportable incident, the manager must
(a) in the case of a reportable incident referred to in section 1.7.1. (1) (a), within 4 hours of the reportable incident, provide a verbal notification of the incident to an inspector, the management co-chair and the worker co-chair of the OHS committee or their designates and the local union,
(b) within 16 hours of a reportable incident, provide a written notice of the incident to an inspector, the management co-chair and the worker co-chair of the OHS committee or their designates and the local union, and
(c) within 60 days of a reportable incident, or within any longer time that may be specified by the chief inspector, submit the

investigation report required under section 1.7.4 (2) to the chief inspector.

(2) The manager must engage with First Nations that may be affected by reportable incidents to determine which reportable incidents must be reported to the First Nations.

Release of a Scene of a Reportable Incident

1.7.3 Subject to any order under section 15 of the Act, the manager must ensure that, except for the purpose of saving life or relieving human suffering, the scene of a reportable incident is not disturbed without approval of

(a) the OHSC worker representative,
(b) an inspector, or
(c) in the case of an incident resulting in loss of life, the chief inspector.

Investigation of a Reportable Incident

1.7.4 (1) The manager must ensure that an investigation into every reportable incident is carried out by persons knowledgeable in the type of work and in collaboration with the management co-chair and the worker co-chair of the OHS committee or their designates.

(2) On completion of the investigation, the manager must prepare a report, signed by all participants listed in subsection (1), that includes the following:

(a) a description of the incident;
(b) a description of the work conducted at the time and place of the incident;
(c) an identification of the causes of the incident and any unsafe conditions, acts, or procedures which may have contributed to the incident;
(d) an identification of any corrective actions that are necessary to prevent the recurrence of similar incidents, including a projected timeline for implementation of the corrective actions;
(e) any additional information required by the chief inspector.

Personal Protective Equipment

Manager's Responsibility

1.8.1 The manager must ensure that

(a) except for protective footwear and prescription eyeglasses, personal protective equipment and safety devices are supplied to persons as appropriate to the task or occasion,
(b) persons are trained in the use of the equipment and devices, and
(c) the equipment and devices are adequate for their purpose.

Responsibility of All Persons

1.8.2 A person must

(a) use, as instructed by a qualified person, the personal protective equipment and safety devices supplied to the person as set out in section 1.8.1, and

(b) ensure the equipment and devices are in good working order.

Protective Hat

1.8.3 A protective hat complying with the relevant requirements of CSA Standard Z94.1-M 1977 "Industrial Protective Headgear," and suitable for the type of work or activity being performed, shall be worn by persons where there is a risk of head injury or where required by the manager or an inspector. Where conditions may cause the hat to be accidentally dislodged

(1) a device shall be incorporated into the hat and worn in order to prevent the hat being dislodged during a fall, and

(2) in the case of underground work, a chin strap and hearing protection muffs attached to the hat shall be worn.

Protective Footwear

1.8.4 Protective footwear complying with the relevant requirements of CSA Standard Z195-M1984 "Protective Footwear", and suitable for the type of work or activity being performed, shall be worn by persons where there is a risk of foot injury or where required by the manager or an inspector.

Eye Protection

1.8.5 Properly fitting goggles, face shields, or other eye protective equipment complying with the relevant requirements of CAN/CSA Standard Z94.3-M88 "Industrial Eye and Face Protectors," and suitable for the type of work or activity being performed, shall be worn by a person who

(1) is handling or is exposed to any material which is likely to injure or irritate the eyes,

(2) is engaged in any work in which there is a risk of eye injury,

(3) has 20/200 vision or is blind in either eye, or

(4) is working or passing through an area which the manager or inspector has designated as requiring such protection.

Contact Lenses

1.8.6 No person shall wear contact lenses where prohibited by the manager.

Hearing Protection

1.8.7 Where noise levels exceed the requirements of Table 2-2, Part 2, persons shall wear hearing protectors which are selected, maintained and used in accordance with CSA Standard Z94.2-94 "Hearing Protectors."

Self Rescuer

1.8.8 All persons going underground in a mine shall carry a self rescuer approved in accordance with NIOSH 42 CFR Part 84 on their person, or in the case of equipment operators maintained within arm's reach while they are operating the equipment.

1.8.9 Notwithstanding section 1.1.4, standards identified in sections 1.8.3, 1.8.4, 1.8.5, 1.8.7, and 1.8.8, shall not be amended unless so directed by the chief inspector.

Workplace Conditions

General

1.9.1 The manager shall

(1) take all reasonable and practicable measures to ensure that the workplace is free of potentially hazardous agents and conditions which could adversely affect the health, safety, or well-being of the workers,

Controls

(2) where practicable, institute controls at the source to ensure that workers are not exposed to a level of any physical, chemical, or radiation hazard in excess of the limits prescribed in the code or by an inspector, with the exception of unusual short term or emergency situations, and

(3) require that persons wear effective personal protective equipment in any situation where control at the source, as required by section 1.9.1 (2), is impractical.

Health and Safety Reports

Monthly Reporting

1.9.2 At the end of each month, the manager must provide

(1) a report of all reported first aid cases to the OHSC and local union or worker representative, and

(2) a report of all reported medical aid cases to the Chief Inspector, the OHSC and local union or worker representative.

Annual Reporting

1.9.3 By January 31 of each year, the manager shall forward to the chief inspector a report for the previous calendar year which includes

(a) the total hours worked at the mine by all mine employees,

(b) the number of lost time injuries,

(c) the number of occasions where employees received medical aid,

(d) the number of days lost, and
(e) where required by an inspector a contractor shall provide the information required under 1.9.3 (a) to (d).

Employees' Right to Refuse Work

Unsafe Work

1.10.1 A person shall not carry out any work or operate any equipment, tool, or appliance if the person has reasonable cause to believe that to do so would create an undue hazard to the health or safety of any person.

1.10.2 A supervisor shall not knowingly perform or permit a worker to perform work which is, or could create, an undue hazard to the health or safety of any person.

Right to Refuse

1.10.3 A person who refuses to carry out any work or operate any equipment, tool, or appliance, in compliance with section 1.10.1, shall immediately report the circumstances to the person's supervisor.

Supervisor Investigates

1.10.4 The supervisor receiving a report under section 1.10.3 shall investigate the matter and ensure that any hazardous condition is remedied without delay; or if, in the supervisor's opinion the report is not valid, the supervisor shall inform the person who made the report.

Investigation

1.10.5 If the procedure provided for in section 1.10.4 fails to resolve the issue and the person continues to refuse to carry out the work, the supervisor or other management representative shall forthwith make an investigation in the presence of the person who made the report, together with another person having knowledge of the work in question and who is

(1) a worker representative or designate of the OHSC if available, or
(2) designated by the local union to represent the person refusing to carry out the work, or
(3) a co-worker selected by the person refusing to carry out the work.

Use of Alternate Worker

1.10.6 If the person still refuses to carry out the work after the person's supervisor and the other person in accordance with section 1.10.5, have investigated the issue and are both of the opinion that no undue hazard exists and that

(1) the refusal is considered to be justifiable for reasons peculiar to that particular person, and
(2) there is no justification for an alternate person to refuse to carry out the work in question then, the supervisor, after informing the alternate

person of the reason for the refusal, may have the alternate perform the work.

Manager Investigates

1.10.7 If the procedures in sections 1.10.4, 1.10.5 and 1.10.6 fail to resolve the issue, the manager shall

(1) conduct an investigation and either develop a plan that is acceptable to the persons who will do the work and which will allow the work to proceed safely, or suspend further work, and

(2) if the work is suspended or allowed to proceed, submit a report to the OHSC, local union, and an inspector, that describes the incident, shows compliance with the code and describes any remedial actions taken.

Training

Training

1.11.1 The manager shall ensure that

(1) workers are adequately trained to do their job or are working under the guidance of someone who has competency both in the job and in giving instruction, and

(2) ensure that all employees receive thorough orientation and basic instruction in safe work practices.

1.11.2 The manager shall maintain a record of all training workers and supervisors have received, and make this record available to an inspector upon request.

Supervision

Open Pit

1.12.1 (1) The manager shall ensure that every worker employed in the mining activity of an open pit mine is under the supervision of

(a) the holder of an open pit shiftboss certificate where more than 6 persons are employed, or

(b) in the case of an open pit mine employing 6 persons or fewer, the holder of a supervisor certificate unless a shiftboss certificate is required by the inspector.

(2) If the manager is the supervisor, the manager must also hold a supervisor's certificate.

Underground Non Coal

1.12.2 (1) The manager shall ensure that every worker working underground in a mine, other than a coal mine, is under the supervision of the holder of an underground shiftboss certificate.

Underground Coal Mine

1.12.3 The manager of an underground coal mine shall ensure that

(1) every worker employed underground is under the supervision of the holder of an underground coal mine fireboss certificate, and

(2) where more than 50 workers per shift are employed underground, a supervisor holding a second class certificate of competency shall be appointed as mine captain/general foreperson and this person shall supervise all underground operations.

1.12.4 A certificate of competency of any class includes that of a lower class.

1.12.5 The manager of an underground coal mine, if required by the chief inspector, shall hold one of the following valid certificates of competency:

(1) an underground coal mine fireboss if fewer than 10 persons are employed underground,

(2) a second class certification of competency if fewer than 30 but more than 9 persons per day are employed underground, and

(3) a first class certificate of competency if 30 or more persons per day are employed underground.

Exploration Supervisor's Certificate

1.12.6 (1) The manager of an exploration site shall ensure that every worker employed in exploration activities is under the supervision of the holder of a supervisor's certificate.

(2) If the manager is a supervisor, the manager must also hold a supervisor's certificate.

Other Work Areas

1.12.7 (1) The manager shall ensure that every worker, other than those persons referred to in sections 1.12.1 (1), 1.12.2, 1.12.3 and 1.12.6, and other than office employees, is under the supervision of the holder of a supervisor's certificate.

(2) If the Manager is a supervisor, the manager must also hold a supervisor's certificate.

Risk of Drowning

1.12.8 The manager must ensure that every worker working where drowning may be a risk is under the supervision of a qualified person carrying out the safe work plan as set out in section 3.3.3.

Certification Procedure

Board Grants Certification

1.13.1 Examinations of applicants for underground shiftboss and fireboss certificates, open pit shiftboss certificates, and certificates of competency shall be carried out by a Board of Examiners, consisting of the chief inspector as chair and two other inspectors appointed by the chief inspector.

1.13.2 One member of the Board of Examiners shall hold first class certificates of competency when dealing with matters pertaining to the certification of candidates for underground coal mines.

1.13.3 The Board of Examiners shall issue a shiftboss, fireboss or competency certificate when satisfied that the applicant

(1) is conversant with the English language,

(2) is the holder of a first aid certificate to a standard required by the chief inspector,

(3) has an adequate knowledge of the *Mines Act*, the regulations, and the code and of other information and material considered necessary by the Board of Examiners, including, but not limited to, mine safety, mine rescue, and blasting practices and procedures, and

(4) has had at least three years' relevant experience in and about the working of a mine, or has obtained a degree or diploma, approved by the Board of Examiners, in engineering from a university or technical institute and has had not less than one year of experience and such experience shall be acceptable to the Board.

1.13.4 Notwithstanding section 1.13.3, the Board of Examiners may grant a restricted shiftboss or fireboss certificate where the type of mine operation would not require the applicant to hold all of the qualifications included in section 1.13.3.

1.13.5 The Board of Examiners may grant a certificate to an applicant who holds a similar certificate granted by a jurisdiction outside the Province, if the Board of Examiners is satisfied that the standard of training and examination is equivalent to that required for the granting of a corresponding certificate under the code and, after oral examination of the applicant, that the applicant is sufficiently qualified.

1.13.6 Notwithstanding sections 1.13.3 and 1.13.4, the manager may issue on a one time basis a provisional shiftboss or fireboss certificate valid for no more than six months to an employee who has acceptable experience and training.

Validity of Certificates

1.13.7 A shiftboss or fireboss certificate or a certificate of competency issued under Part 1 of the code is conditional on the holder maintaining in force the other certificates required under section 1.13.3 (2), or as required by the Board of Examiners.

1.13.8 Repealed.

Suspension of Shiftboss, Fireboss or Competency Certificate

1.13.9 (1) A shiftboss or fireboss certificate or certificate of competency issued under this Part may be suspended by an inspector if the holder fails to comply with the Act, the regulations, or this code, or is by reason of incompetence or negligence, unfit to carry out their duties.

(2) The inspector shall immediately inform the Board of Examiners of any suspension under subsection (1), and the Board may
(a) conduct an investigation,
(b) interview the holder of the certificate, and
(c) vary the suspension or cancel the certificate

(3) A person whose certificate is cancelled or suspended under this section may appeal the decision to the chief inspector.

Supervisor Certification

1.13.10 (1) The chief inspector shall issue a supervisor's certificate to an applicant who has passed an examination on the sections of the *Mines Act*, the regulations, and this code.

(2) The certificate is valid for five years from date of issuance.

Suspension of Supervisor's Certificate by Inspector

1.13.11 (1) A supervisor's certificate issued under this Part may be suspended for a stated period of time by an inspector if the holder fails to comply with the Act, the regulations or this code, or is by the reason of incompetence or negligence unfit to carry out their duties.

(2) The inspector shall inform the chief inspector immediately of any suspension under subsection (1), and the chief inspector may
(a) conduct an investigation,
(b) interview the holder of the certificate, and
(c) vary the decision of the district inspector or cancel the certificate.

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Workplace Contaminants

Maximum Allowable Exposures

2.1.1 (1) The manager must ensure that no employee is exposed to airborne concentrations of a substance or noise in excess of the limit specified in

(a) Table 2-1 or 2-2, or

(b) the publication by the American Conference of Governmental Industrial Hygienists entitled, "Threshold Limit Values for Chemical Substances and Physical Agents & Biological Exposure Indices", as amended from time to time, if the substance or noise is not specified in Table 2-1 or Table 2-2.

(2) Repealed.

Shifts Longer Than 8 Hours

2.1.2 If a shift at a mine is longer than 8 hours and involves exposure to an airborne substance that is listed in Table 2-4, other than a substance indicated in the table to be category I-A, I-B or I-C, the manager must ensure that the time-weighted average concentration of the airborne substance is adjusted using the applicable adjustment factor determined as follows:

(a) for substances indicated to be category II in Table 2-4, a daily adjustment factor as determined using the formula

$$F_a = 8/H_d$$

F_a = adjustment factor
H_d = exposure duration in hours per shift;

(b) for substances indicated to be category III in Table 2-4, a weekly adjustment factor as determined using the formula

$$F_a = 40/H_{wk}$$

F_a = adjustment factor
H_{wk} = average duration of work shifts per week based on a repetitive work cycle;

(c) for substances indicated to be category IV in Table 2-4, the lowest of the following two adjustment factors:

(i) a daily adjustment factor as determined using the formula set out in paragraph (a);

(ii) a weekly adjustment factor as determined using the formula set out in paragraph (b);

(d) for substances not listed in Table 2-4, the lowest of the following two adjustment factors:

(i) a daily adjustment factor as determined using the formula

$$F_a = \frac{8}{H_d} \times \frac{24 - H_d}{16}$$

F_a = adjustment factor
H_d = exposure duration in hours per shift;

(ii) a weekly adjustment factor as determined using the formula

$$F_a = \frac{40}{H_{wk}} \times \frac{168 - H_{wk}}{128}$$

F_a = adjustment factor
H_{wk} = average duration of work shifts per week based on a repetitive work cycle.

Monitoring Contaminants

2.1.3 (1) The manager shall establish a written program, in accordance with the chief inspector's publication "Workplace Monitoring Procedures Manual" for a qualified person to monitor workplace contaminants as often as necessary to ensure compliance with section 2.1.1. The program shall specify the substances and locations to be monitored and the frequency of monitoring. The results of the program shall be available for examination by an inspector.

(2) A record of the monitoring program shall be kept on file for the life of the mine and transferred to the chief inspector upon abandonment of the mine.

Oxygen Content

2.1.4 Persons shall not work or remain in any part of a mine where they may be exposed to air with an oxygen content less than 19.5% by volume unless provided with an alternate air supply.

Control of Workplace Hazards

Workplace to Be Hazard Free

2.2.1 The manager shall develop and implement an effective housekeeping program to ensure that all workplaces and travelways are maintained in a safe condition, that materials and equipment are stored in a manner so as not to endanger persons, and that appropriate action is taken whenever necessary to maintain a hazard-free environment.

2.2.2 An inspector may require the manager to satisfy the inspector that it is not practicable to comply with section 1.9.1 (2) with proper engineering controls, and the inspector shall communicate the inspector's decision to the OHSC.

Hazardous Materials & Waste

Asbestos Controls and Substitutes

2.3.1 The manager shall ensure that

(1) as effective, less hazardous, asbestos-free substitutes become available and practical for such items as brake lining, insulation, and fire retardant materials, they shall be used,

(2) where there is no substitute available for an asbestos-containing material, a procedure acceptable to the chief inspector is developed to protect the health of persons using, maintaining, or exposed to such material,

(3) notwithstanding subsection (2), items containing crocidolite asbestos are not to be used at a mine,

(4) spraying of asbestos, or material containing more than 1% of asbestos, is prohibited, and

(5) protective clothing and equipment made of material containing asbestos is used in such a manner as to prevent the release of asbestos fibers.

Disposal of Asbestos

2.3.2 Where any work involves the use, handling, or disposal of asbestos or materials containing asbestos, the manager shall ensure that the procedures outlined in the manual "Safework Practices for Handling Asbestos Revision 10 1996" published by the Board are followed, and that the inspector and the OHSC are notified of the procedures.

Storage of Hazardous Materials

2.3.3 The manager shall ensure that all dangerous or potentially hazardous materials are stored in designated storage areas, acceptable to an inspector, which are well ventilated to prevent any accumulation of dangerous fumes and so arranged that incompatible materials, which could produce a harmful reaction if combined, are adequately separated.

Proper Containers

2.3.4 The manager shall ensure that all dangerous or potentially hazardous materials are stored in containers that are designed, constructed, and maintained in such a manner as to ensure proper containment of their contents under the environmental conditions in which they are stored.

Hazardous Dust

2.3.5 In a building where dust or other material could by becoming airborne cause a hazard by restricting vision, or could cause a fire or explosion or be potentially hazardous if inhaled or ingested, it shall without delay

be removed by suitable means such as vacuuming, wet sweeping, or wet shoveling, or be suppressed.

Hazardous Waste

2.3.6 Waste materials which could be detrimental to a person's health, or could cause a fire or explosion, shall be

(1) stored in impervious containers, labeled to clearly identify the contents and the nature of the hazard, and where the contents are flammable or capable of producing vapours or gases, the containers shall be covered and located in well-ventilated areas, and

(2) collected, handled, sorted, and disposed of by persons trained to safely handle the waste material and wearing any necessary protective clothing and equipment to safeguard their own and any other person's health and safety.

Spills

2.3.7 Spills or releases of hazardous waste materials shall be cleaned up as soon as possible by persons trained to safely handle the waste material and wearing any necessary protective clothing and equipment to safeguard their health and safety.

Flammable Waste Storage

2.3.8 Waste materials containing solvents, grease, paints, or other flammable substances shall be stored in appropriate covered containers while awaiting disposal.

Radiation

2.3.9 The manager shall

(1) ensure that equipment which emits harmful levels of infrared or ultraviolet radiation is shielded to protect employees from exposure to such radiation, and

(2) supply personal protective equipment to any person who could become exposed to harmful radiation levels and the person shall wear the protective equipment.

2.3.10 Notwithstanding section 1.1.4 of the code, the manager shall ensure that employees are not exposed to non-ionizing radiation which exceeds the recommendations contained in the publication "*Safety Code 6, Recommended Safety Procedures for the Installation and Use of Radio Frequency and Microwave Devices in the Frequency Range of 10 MHz to 300 GHz*", published by Health and Welfare, Canada.

2.3.11 (1) A worker shall not be exposed to ionizing radiation to a level greater than any of the following:

(a) 4.0 Working Level Months (20 millisieverts) of radon decay products per year averaged over 5 years, and 10.0 Working Level Months (50 millisieverts) in a single year;
(b) 20 millisieverts (2 Rem) of gamma radiation per year averaged over 5 years, and 50 millisieverts (5 Rem) in a single year;
(c) a combination of radon decay products and gamma radiation totaling 20 millisieverts averaged over 5 years or 50 millisieverts in a single year.

(2) Where there are indicators that uranium or thorium levels are in a grade of 0.05% by weight or greater, the mine manager shall conduct a survey using appropriate calibrated instruments to ensure that worker exposure rates do not exceed those stated in section 2.3.11 (1).

(3) Where measurements indicate a worker is exposed to gamma radiation greater than 2.5 microsieverts per hour (0.25 millirem per hour) at 0.5 meters from the source, a gamma radiation dosimeter of a type acceptable to the chief inspector shall be provided to and worn by each worker who is exposed.

(4) All radioactive materials shall be used, handled, stored, transported, and disposed of in accordance with the Canadian Nuclear Safety Commission Regulations.

(5) The manager shall develop and implement industrial hygiene standards, acceptable to the chief inspector, for use at uranium drill sites.

(6) The manager shall implement the Province of British Columbia Radiation Guidelines for Mineral Exploration.

Spraying of Materials Containing Isocyanates

2.3.12 Where paints, protective coatings, adhesives, or insulating materials containing isocyanate compounds or other compounds that have similar sensitizing effects are being applied, persons shall be protected from the airborne contaminants by

(1) where practicable, performing the work in an isolated enclosure having sufficient exhaust ventilation to ensure that contaminated air is not released from the enclosure into areas where unprotected persons can enter, and

(2) where methylene bisphenyl isocyanate is used, ensuring that a supplied-air respirator (SAR) approved by NIOSH or MSHA is worn by all exposed persons, in accordance with the following concentrations:
(i) up to 2 mg/m^3: SAR or SCBA,
(ii) up to 5 mg/m^3: SAR operated continuous flow, or SCBA,
(iii) up to 10 mg/m^3: positive pressure, full-facepiece SAR,

(iv) up to 100 mg/m^3: positive pressure, full-facepiece SCBA, and

(v) emergency or planned entry into unknown concentrations or immediately dangerous to life or health: positive pressure, full-facepiece SCBA, or positive pressure full-facepiece SAR with an auxiliary positive pressure SCBA.

(3) A respirator type shall be selected in accordance with assigned protection factors as outlined in Table 2 of CSA Standard Z94.4-93.

Harmful Chemical Exposure

Emergency Wash Facilities

2.4.1 Where persons may be exposed to corrosive or other chemicals harmful to the eyes or skin, the manager shall ensure that eye wash equipment, emergency water baths or showers, or other suitable means are immediately available to effectively cleanse the affected body areas.

(1) The emergency shower facility shall provide a tempered supply of potable water for a minimum of 15 minutes.

(2) In cases where it is not practical to install a permanent facility portable eye wash stations and/or portable emergency wash facilities may be installed as applicable.

(3) Provisions shall be taken to ensure that water supplies and eye wash isotonic fluids are protected from freezing and are adequately maintained.

Protection From Contaminants

2.4.2 (1) Where the nature of the work causes a person's clothing or skin to be contaminated with substances which could cause injury to, or endanger the health of, the person or contaminate other areas on or off the minesite, the manager shall provide

(a) shower or wash facilities to enable employees to effectively remove all contaminants, and

(b) separate storage areas for street and work clothing to prevent contamination of the street clothing.

(2) Where contaminated clothing is not to be removed from the minesite, the manager shall provide

(a) a means for cleaning the contaminated clothing, or

(b) suitable protective clothing.

(3) A person exposed to contaminants shall

(a) cleanse any affected skin areas as soon as practicable,

(b) not consume or handle food or tobacco products until the person's hands and face are free of contamination,

(c) ensure that the person removes contaminated clothing and disposes of it, and

(d) not leave the mine at the end of the person's shift until all affected areas of the person's skin have been cleansed of contamination.

Abrasive Blasting

No Silica or Lead

2.5.1 Abrasive blasting material containing 1%, or more, free silica or lead shall not be used.

2.5.2 All abrasive blasting and similar operations, when carried out within a building, shall be conducted in an isolated enclosure to prevent injury to persons. The enclosures shall have sufficient exhaust ventilation to ensure that a continuous inward flow of air is maintained at all openings in the enclosure during the blasting operation.

2.5.3 All abrasive blasting and similar operations, when carried out outside a building, shall be conducted in a place and manner to prevent injury to persons.

Protective Equipment

2.5.4 (1) Where an abrasive blasting operation is being carried out, the manager shall supply and employees shall wear air-supplied hoods or respirators suitable for the work, together with gloves, leggings, and clothing designed to protect the employees from dust, and projected abrasive or other material.

(2) Air supplied to the hood or respirator shall meet the requirements of CSA Standard CAN3-Z180.1-M85 "Compressed Breathing Air and Systems," as updated from time to time, and the volume of air supplied shall be sufficient for respiration and to prevent the entry of contaminants into the hood or respirator, and shall not be less than 105 L per minute at the mask.

Nozzles and Valves

2.5.5 (1) Blast cleaning nozzles shall be equipped with an operating valve which must be held open manually, and the normal operation of this valve shall not be defeated. A support shall be provided on which the nozzle can be secured when not in use.

(2) In addition to the operating valve required by subsection (1), another operating control shall be readily accessible to the operator to enable the flow of abrasive to be stopped immediately.

Workers to be Removed

2.5.6 Where an abrasive blasting or a similar operation releases harmful substances to the atmosphere, persons who are not required to assist in the operation shall be removed from contaminated areas. Where such removal is not practical, the exposed persons shall be advised of the hazard and supplied with suitable personal protective equipment which they shall wear.

Noise

Mufflers Required

2.6.1 Any machinery or equipment which, when operating, exposes the operator or persons in the vicinity to noise levels in excess of those prescribed in Table 2-2, Part 2, for unprotected ears, shall, if practicable, be fitted with a properly maintained muffler or other noise reducing device.

Use of Lasers

Laser Standard

2.7.1 Where laser emitting devices are used, they shall be classified and used in accordance with the procedures outlined in ANSI Z136.1 - 1993 – "American National Standard for the Safe Use of Lasers".

Protection Required

2.7.2 Warning signs shall be posted and properly constructed screens or enclosures used wherever there could be a danger to any person from an inadvertent exposure to a laser beam.

Illumination

Lighting Standards

2.8.1 The manager shall ensure that at all working places, suitable and adequate illumination is provided meeting the standards set out in the ANSI/IES Standard RP-7- 1991: "American National Standard Practice for Industrial Lighting", as updated from time to time, unless otherwise authorized by the code.

Surface Illumination

2.8.2 The manager shall ensure that there is a separate and independent emergency source of illumination at all places where a hazard could be caused by a failure of the normal lighting system, and the emergency lighting system shall

(1) where it is a part of a permanent installation, turn on automatically when the normal lighting fails,

(2) provide adequate illumination to allow employees to initiate emergency shutdown procedures and leave their work areas safely, and

(3) be tested as frequently as necessary to ensure that it will function when required.

Underground Illumination

2.8.3 The manager shall have suitable permanent lighting installed in an underground mine to provide adequate illumination in the following locations

(1) all workshops, service garages, and other places where moving machinery or equipment could be a hazard,

(2) main shaft stations and active shaft landings,

(3) first aid stations, and

(4) conveyor galleries, drives, and transfer stations.

Cap Lamps to be Provided

2.8.4 The manager shall provide every person entering an underground mine with an approved cap lamp and the person shall keep the lamp in the person's possession while the person is underground.

Cap Lamps Specifications

2.8.5 All cap lamps, shall be capable of providing a peak luminance of at least 1500 lux at 1.2 m from the light source, throughout a working shift.

2.8.6 The manager of an underground mine shall develop a procedure for assessing and maintaining cap lamps and a copy of the procedure and the results of the assessment shall be made available to an inspector.

Auxiliary Lighting

2.8.7 Where a person has to assess ground conditions in an underground mine at a distance greater than the effective range of a cap lamp, the manager shall supply, and the person shall use, auxiliary lighting adequate for safely carrying out the assessment.

Musculoskeletal Disorders

2.9.1 Where the equipment, work procedure, or working condition in a work area has caused injurious inflammation of muscles, tendons, or bursae of the upper limbs of the persons doing the work, and it is demonstrated to be from repetitive or forceful use, the chief inspector shall, where practicable, require implementation of one or more of the following preventive measures

(1) modification of work procedures or equipment to reduce physical demands on affected body areas, or

(2) a rescheduling of work to permit safe adjustment to unaccustomed task requirements.

Thermal Environment

Heat or Cold Stress

2.10.1 Where it is not reasonably practical to control thermal conditions, and the nature of the work can cause distress or illness to a person, the manager shall institute a program to

(1) instruct employees in the possible adverse effects of their working environment,

(2) instruct employees how to recognize symptoms of heat or cold stress and what emergency treatment should be applied, and

(3) monitor thermal conditions to identify when persons could be adversely affected by heat and cold stress, and if protective measures are required to adequately protect persons, the manager shall advise an inspector of the measures taken.

Lunchrooms & Sanitary Conveniences

2.11.1 The manager shall provide a source of cool, potable drinking water complying with the Safe Drinking Water Guidelines of the Ministry of Health in locations that

(1) are reasonably accessible to employees,

(2) are kept clean and in a sanitary condition, and

(3) are designed to permit the water to be dispensed and drunk in a sanitary manner.

Use of Solder

2.11.2 New installations of pipes and vessels, and changes to existing pipes and vessels which carry water to be used in whole or in part by persons for drinking purposes, shall not be constructed using solder containing more than 1% lead.